



COMP 5422 Lecture 4

Multimedia Computing, Systems and Applications

Basics of Image

Spring 2026

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**This lecture contains
3 parts**

1. Color
2. Digital image
3. Lossless predictive coding (LPC) for image

Learning Outcomes

1. Color

- Understand the concept of color and visible spectrum.
- Some popular color models such as RGB, CMY and HSI.
- Convert an RGB color to a CMY color or an HSI color.

2. Digital image

- Understand the production and acquisition of a digital image and the related concepts such as resolution or bit depth.
- Extract bit plane for a gray image or a color channel.

3. Lossless predictive coding (LPC) in image

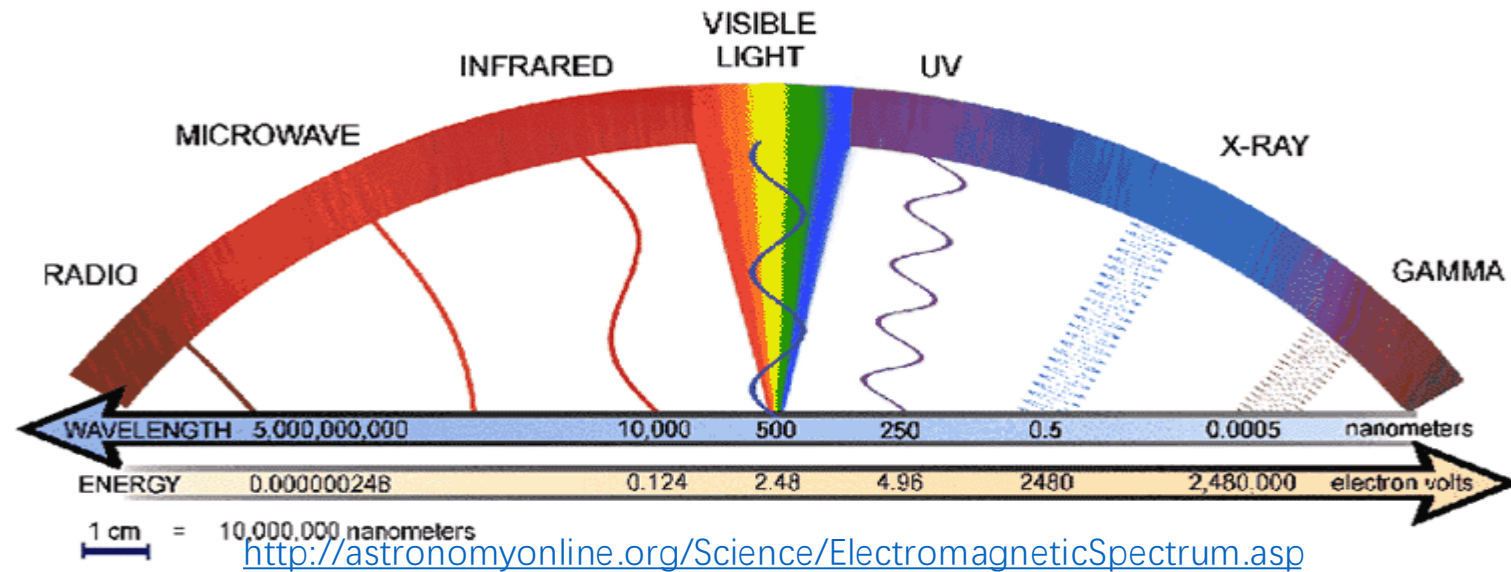
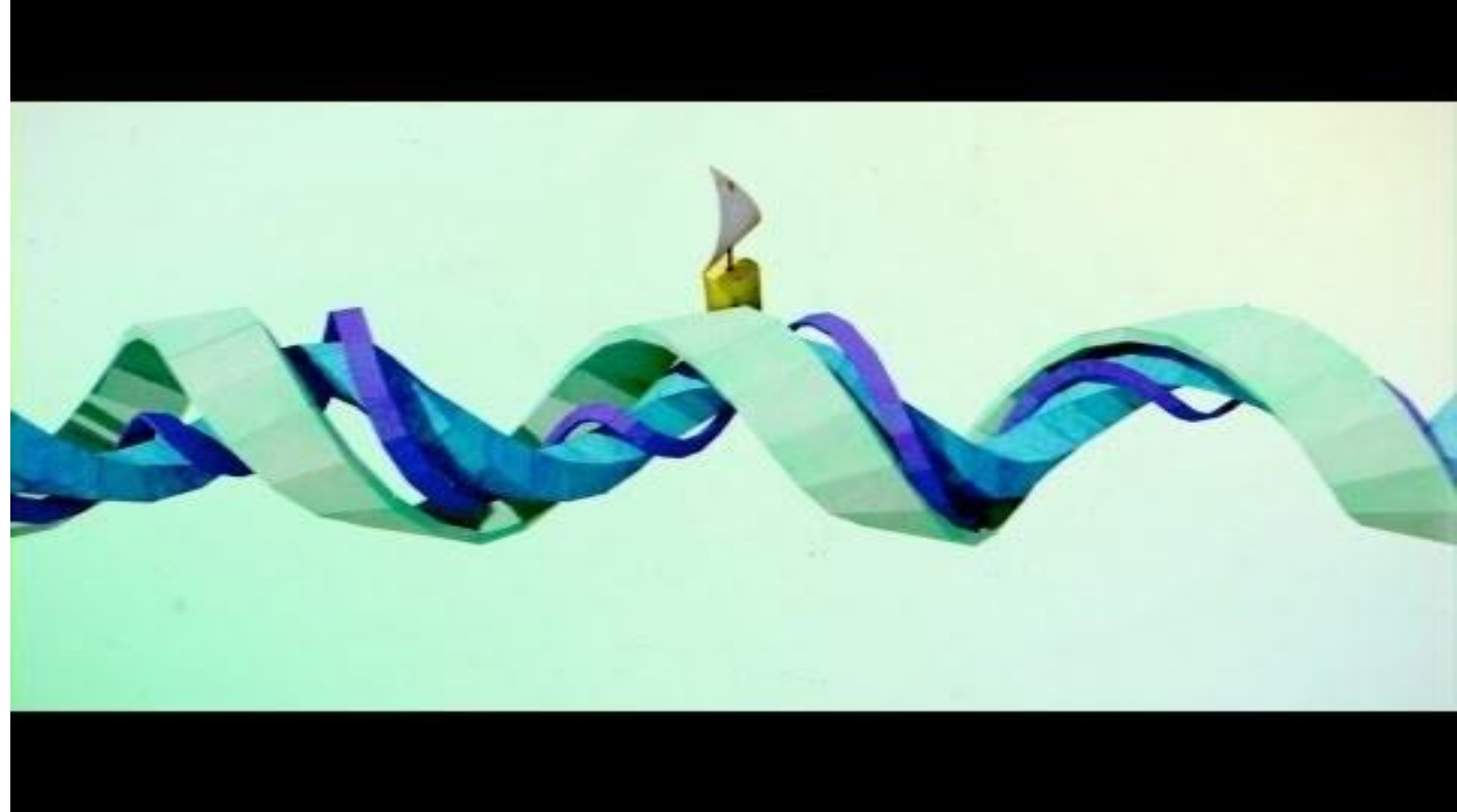
- Perform lossless predictive coding for an image.

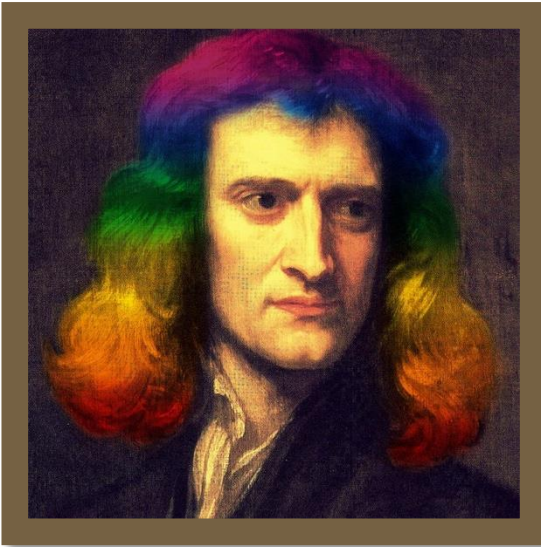
Color Model

- Color fundamentals
- Color models and conversions
 - RGB, CMY, HSI

What is color?

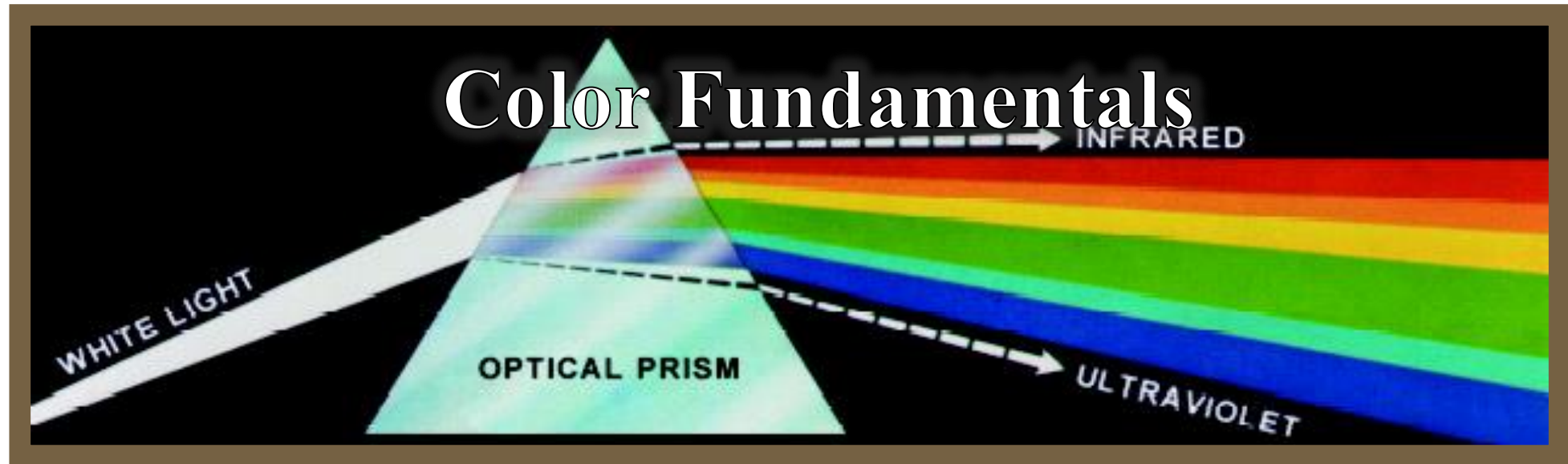
- Color perceived from an object is determined by the nature of light **reflected** from that object. An object that favors reflectance in a limited range of the **visible spectrum** exhibits a specific color.
- The visible spectrum is the portion of the electromagnetic spectrum that is visible to the human eye.
- A white object reflects light that is **balanced** in all visible band.





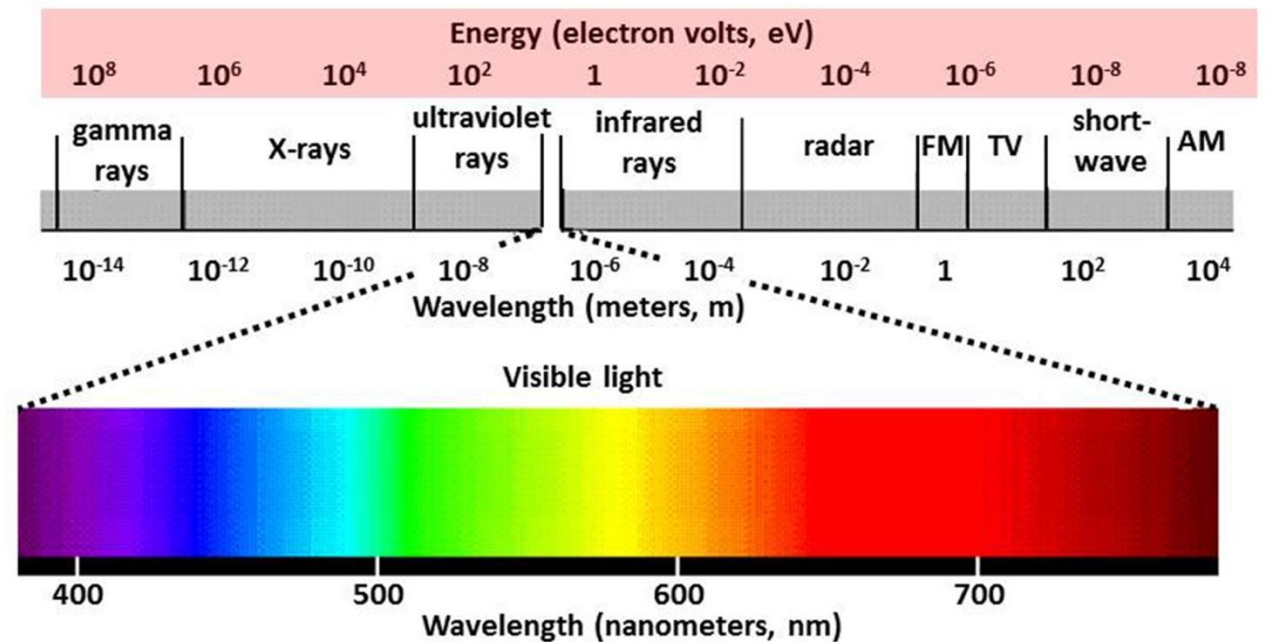
<https://99designs.hk/blog/tips/why-are-there-7-colors-rainbow/>

- In 1666 **Newton** discovered that a beam of sunlight passed through a **prism** will break into a **spectrum** of colors ranging from violet at one end to red at the other.
- **No** color in the spectrum ends abruptly.



Achromatic and Chromatic light

- Achromatic light (without color) is described only by intensity (the amount of gray level).
 - Achromatic light is what you see on a black/white TV.
 - Gray level is a scalar measure of intensity that ranges from black, to grays and to white.
- Chromatic light spans the electromagnetic spectrum from approximately 400nm to 700nm.



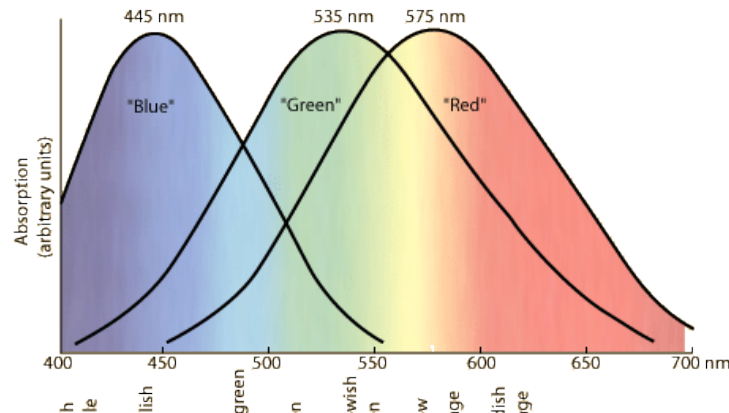
The Dress



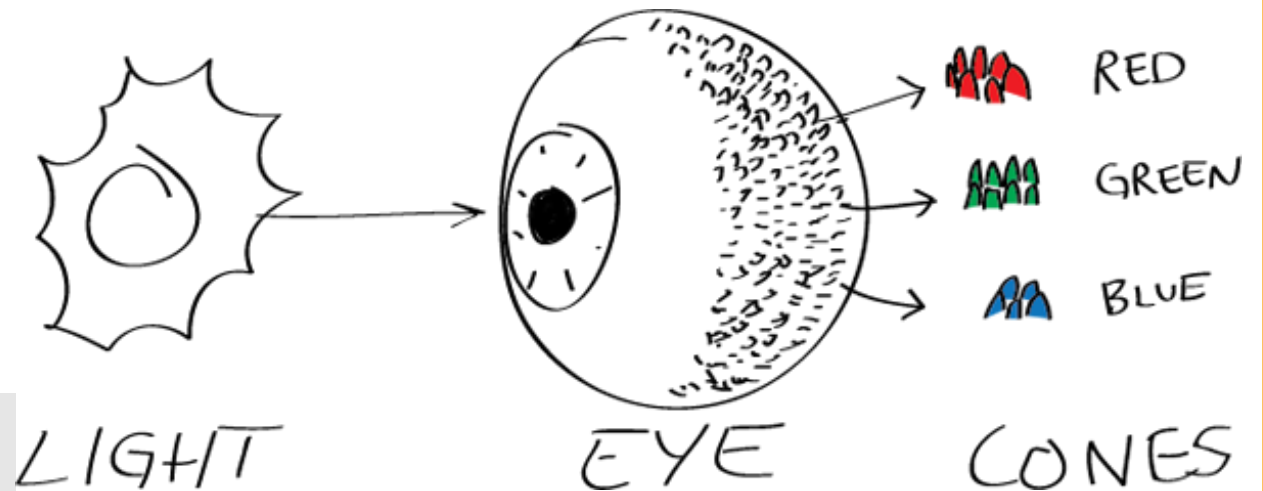
Color is a highly psychological concept!

Primary Colors

- Detailed experimental evidence has established that the 6~7 million cones in the human eye can be divided into **three principal sensing categories: red, green and blue**.
- Due to the absorption characteristics of human eyes, colors are perceived as variable combinations of primary colors **R, G and B**.

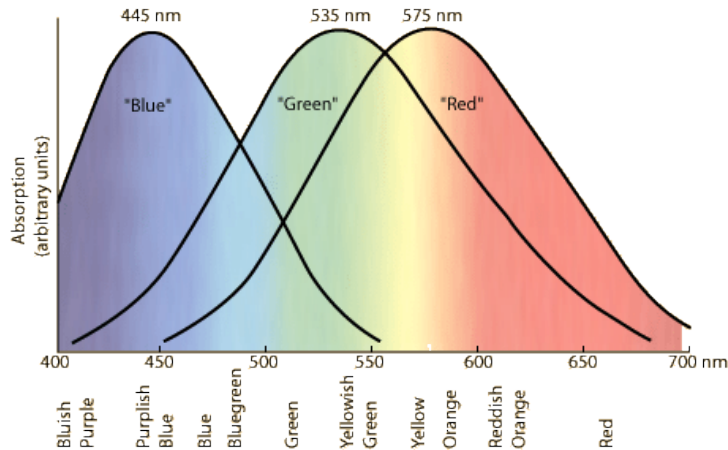


Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

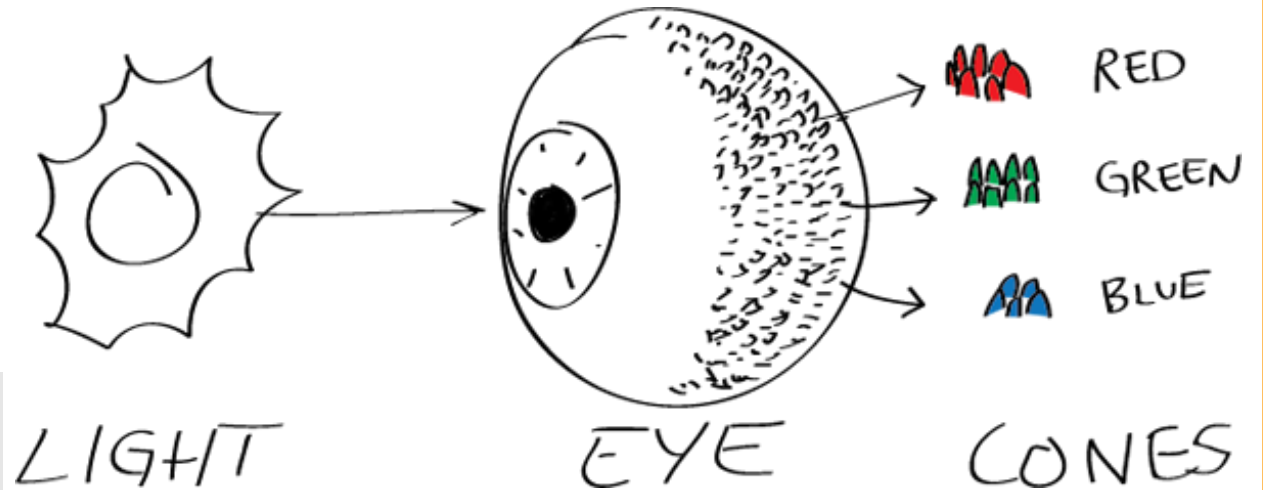


Primary Colors

- Having three specific primary color wavelengths for the purpose of **standardization** does **NOT** mean that RGB acting alone can generate all spectrum colors.
- Use of the word “**primary**” has been widely misinterpreted to mean that RGB can produce all visible colors.

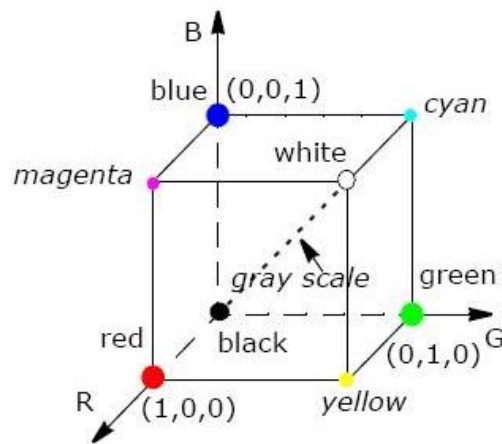


Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.



Color Models

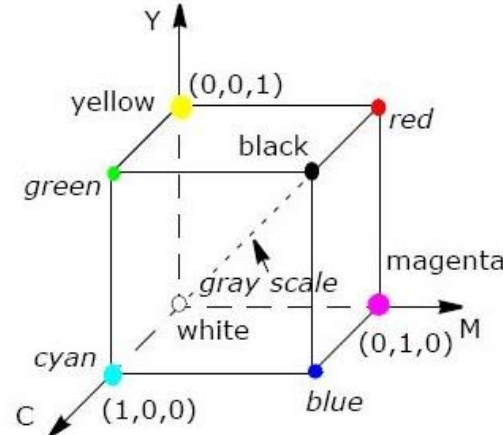
- Color models: used to specify colors in a standard way. Color model is a coordinate system where each color is represented by a single point.



RGB

Red, green, blue

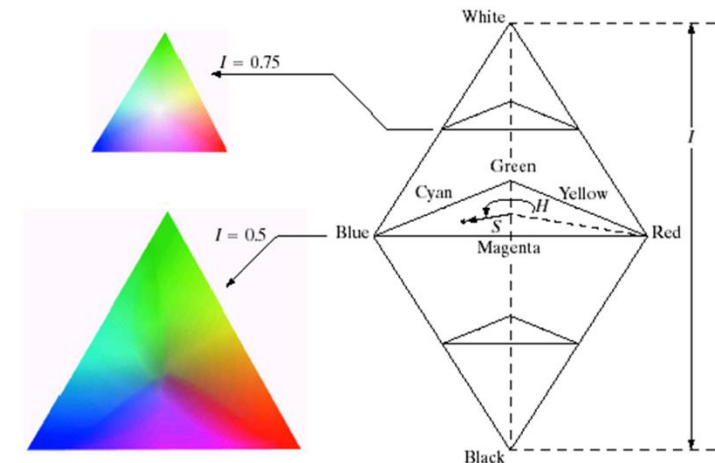
Each color appears in its **primary** spectral **components** of red, green and blue.



CMY (for printing)

Cyan, magenta, yellow

Each color appears in its secondary colors of light, Cyan, magenta and yellow.

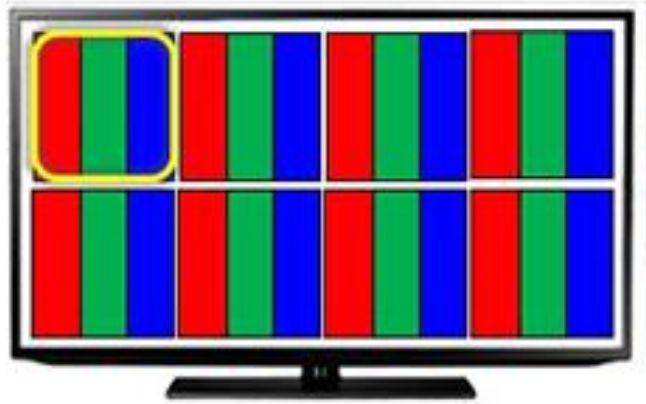


**HSI (sue for 图
像分析)**

Hue, saturation, intensity
Each color is specified by its brightness (I) and color information (H, S). It is close to **human visual system** and **makes** color description easier.

Color Models

- Each model is oriented toward a hardware or application.



RGB

Red, green, blue

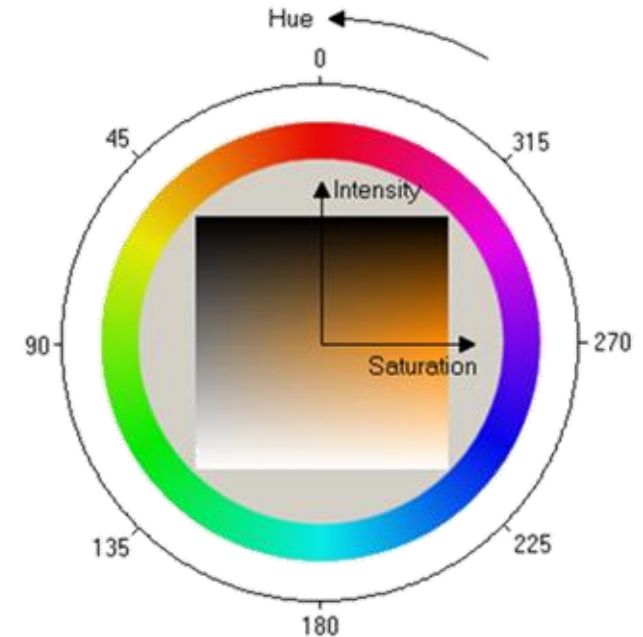
Color monitors, cameras, color
image processing



CMY

Cyan, magenta, yellow

Color printing



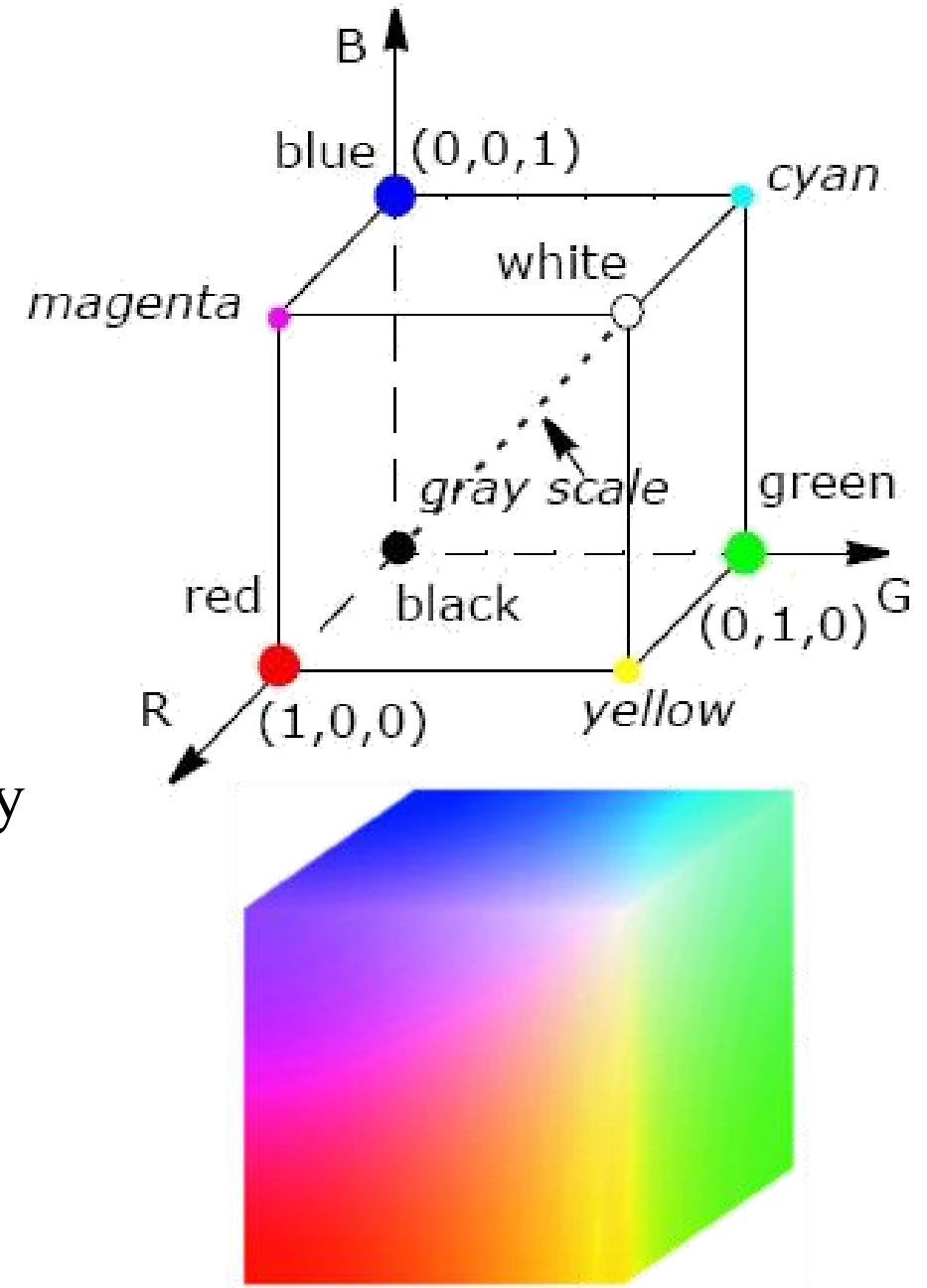
HSI

Hue, saturation, intensity

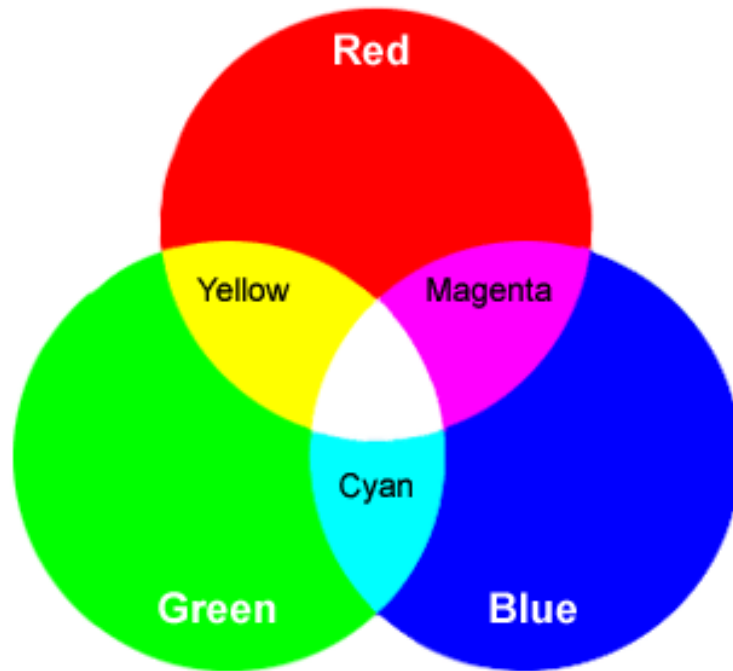
Broadly used in color image
processing

RGB

- Each color appears in its **primary components** of red, green and blue.
- Different colors are points inside the cube.
- The main purpose of the RGB color model is for the sensing, representation and display of images in electronic systems, such as televisions and computers.
- RGB color model has a solid theory behind it, based in human **perception of colors**.



Secondary Colors

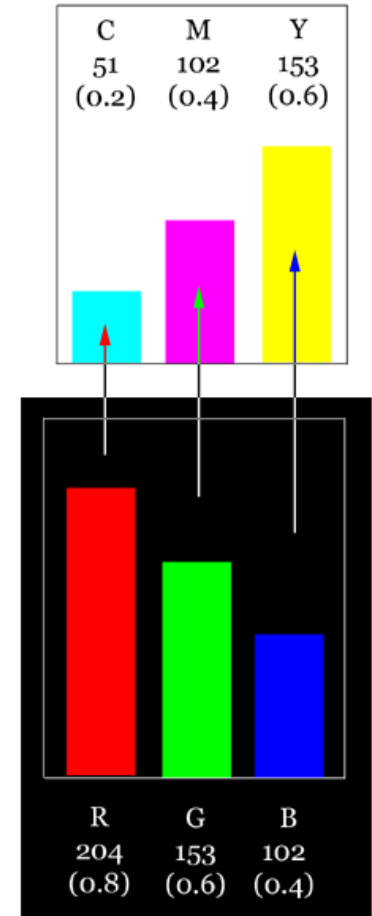
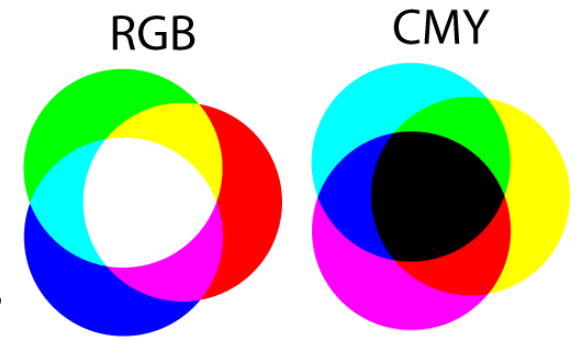
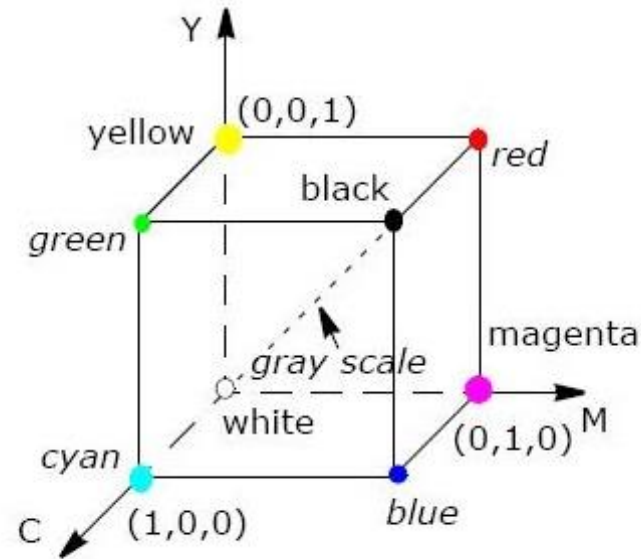


- Primary colors can be added to produce secondary colors:
 - **Magenta**: (red plus blue, opposite to green)
 - **Cyan**: (green plus blue, opposite to red)
 - **Yellow**: (red plus green, opposite to blue)

CMY

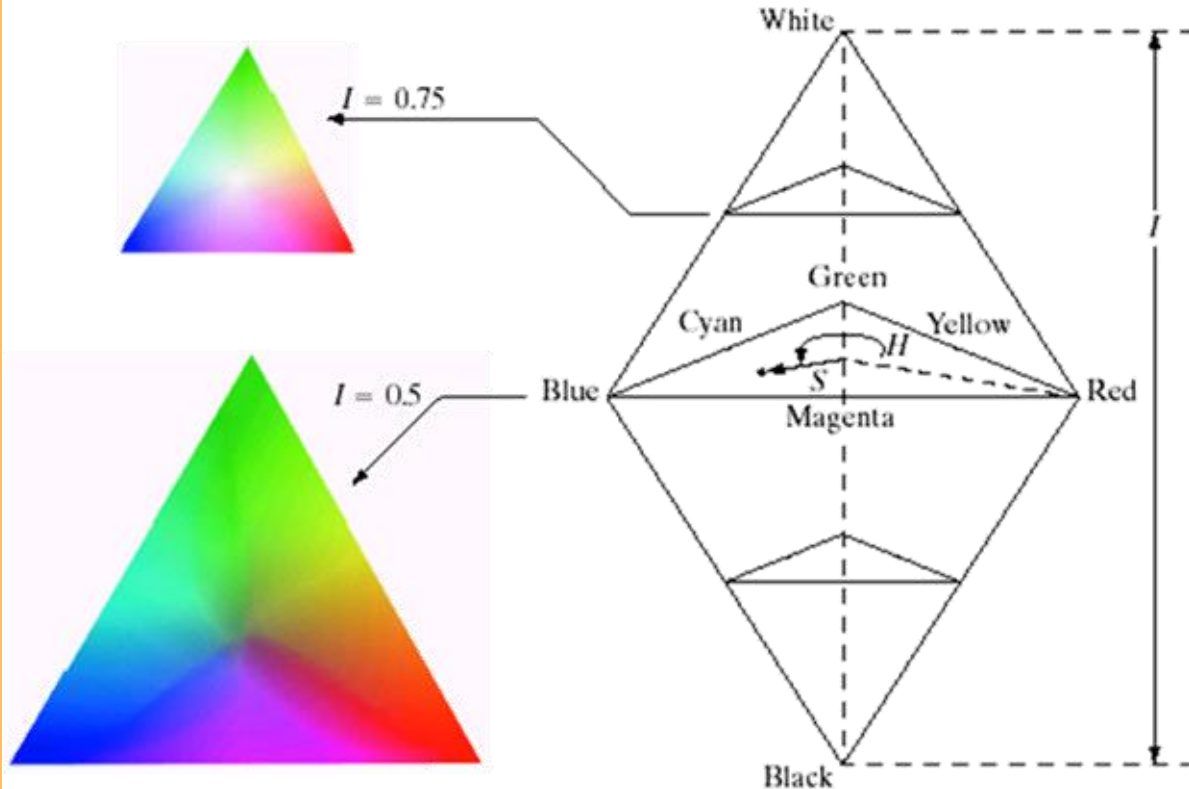
- Cyan, magenta and yellow are the secondary colors of light, or the primary colors of **pigments**.

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

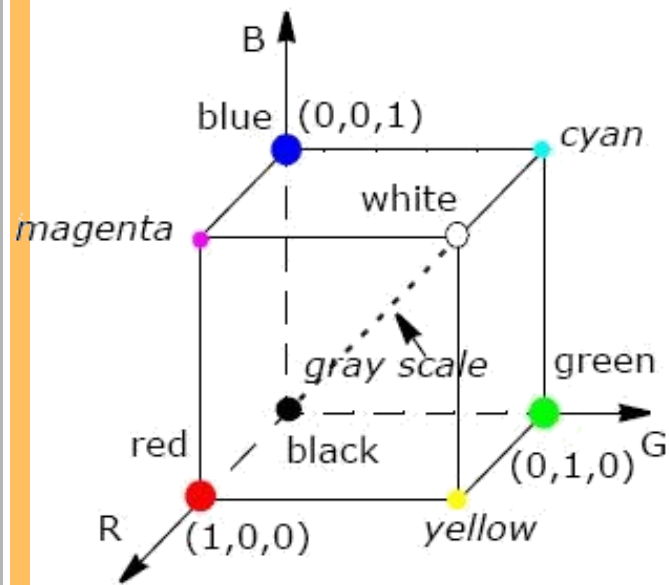


- Most devices that deposit colored pigments on **paper** (color printers and copiers) require CMY data input.
- Pure cyan does not reflect red; pure magenta does not reflect green; pure yellow does not reflect blue.

HSI



- **Intensity**: the most useful descriptor of gray images, distinguishes gray levels.
- **Hue** is associated with the **dominant wavelength** (color) of the **pure color** observed in a mixture of light waves. It represents the **dominant color** perceived by an observer.
 - Distinguishes red, yellow, green, etc.
 - When we call an object red or yellow, we are specifying its hue.
- **Saturation**: relative **purity** or the amount of white light mixed with a hue.
 - The **pure** spectrum colors are **fully** saturated.
 - E.g. pink is less saturated than red.



RGB

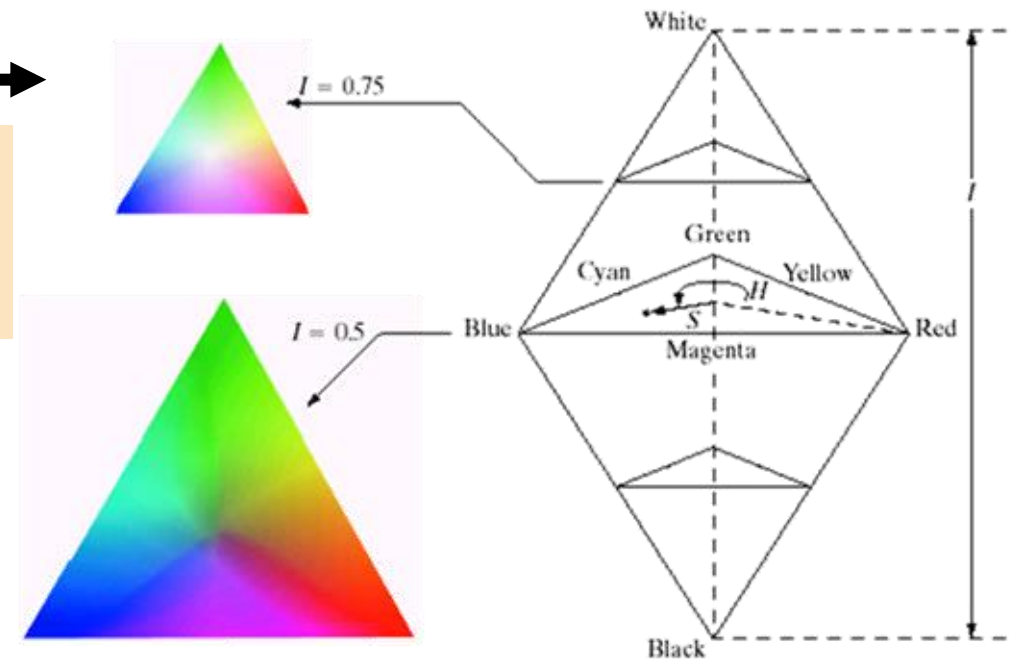
$$\theta = \cos^{-1} \left\{ \frac{[(R - G) + (R - B)]/2}{[(R - G)^2 + (R - B)(G - B)]^{1/2}} \right\}$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

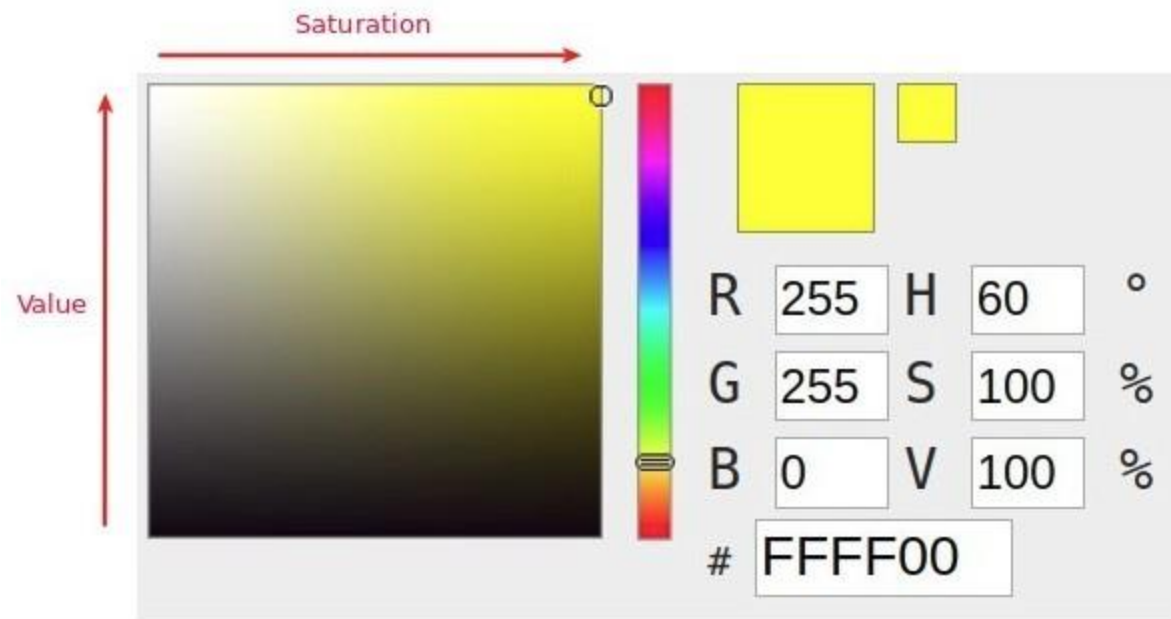
$$I = (R + G + B)/3$$

Similarly, an RGB color can be converted to an HSI color.



HSI

- A slice of the cylinder when $H=60$



HTML Color Codes

- Format: Each HTML code contains symbol “#” and 6 letters or numbers. These numbers are in hexadecimal numeral system. For example, "FF" in hexadecimal represents number 255 in Decimal.
- The first two symbols in HTML color code represents the intensity of **red** color. 00 is the least and FF is the most intense. The third and fourth represent the intensity of **green** and the fifth and sixth represent the intensity of **blue**.

Orange colors

#FF7F50

Color	HTML / CSS Color Name	Hex Code #RRGGBB	Decimal Code (R,G,B)
	coral	#FF7F50	rgb(255,127,80)
	tomato	#FF6347	rgb(255,99,71)
	orangered	#FF4500	rgb(255,69,0)
	gold	#FFD700	rgb(255,215,0)
	orange	#FFA500	rgb(255,165,0)
	darkorange	#FF8C00	rgb(255,140,0)

<https://www.rapidtables.com/web/color/html-color-codes.html>

Digital Image

- Definition, sensing and acquisition
- Representation (bit depth, resolution)

What is a digital Image

- A two-dimensional function $f(x, y)$ where (x, y) are spatial coordinates and the amplitude f is called the intensity or gray level.
- When x , y and f are discrete quantities, the image is called a digital image.

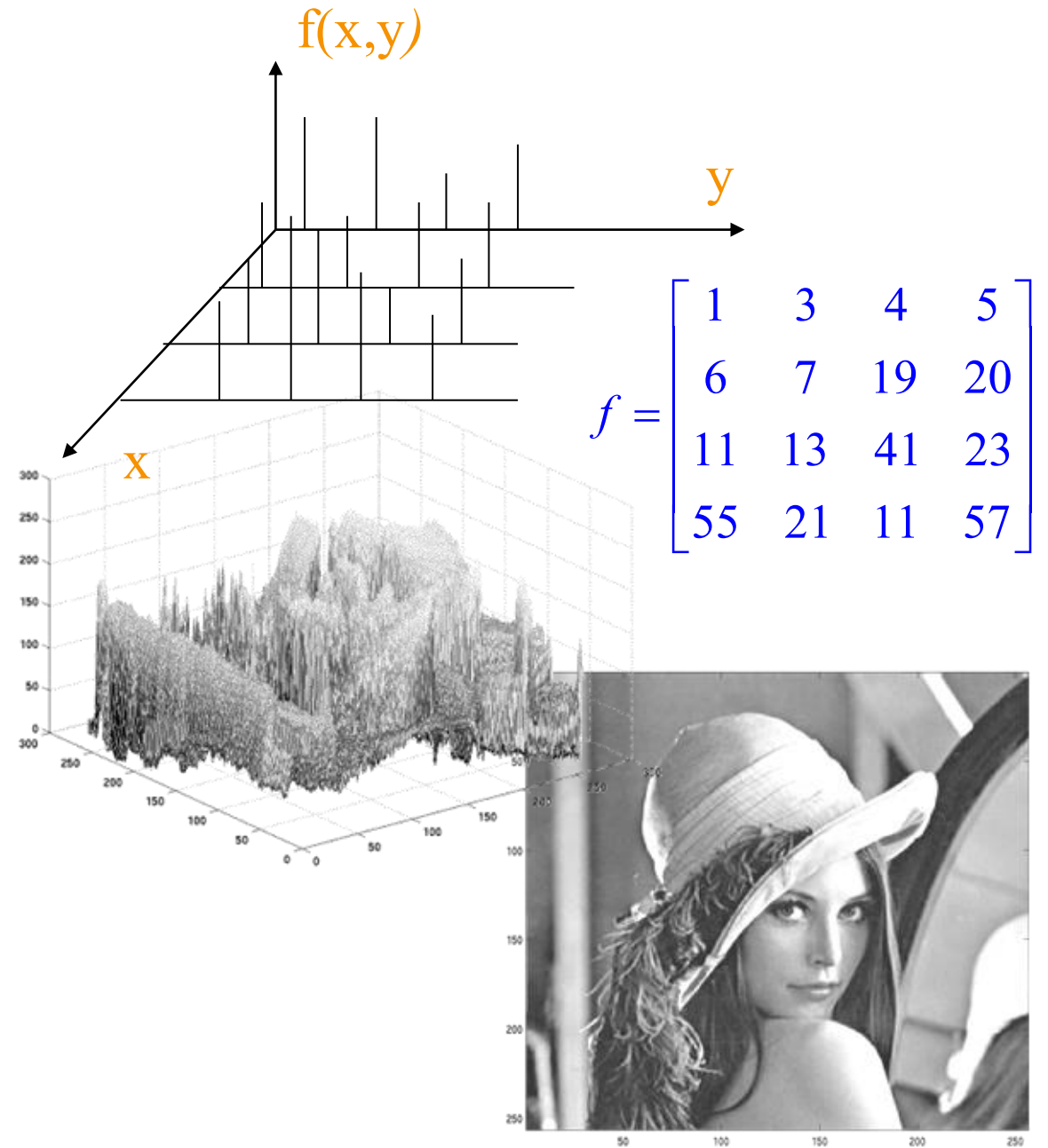


Image Sensing and Acquisition

- Image can be generated by energy of the **illumination source** reflected (natural scenes) or transmitted through objects (e.g., X-ray).
- A **sensor** detects the energy and converts it to electrical signals.
- Sensor should have a material that is **responsive** to the particular type of energy being detected.

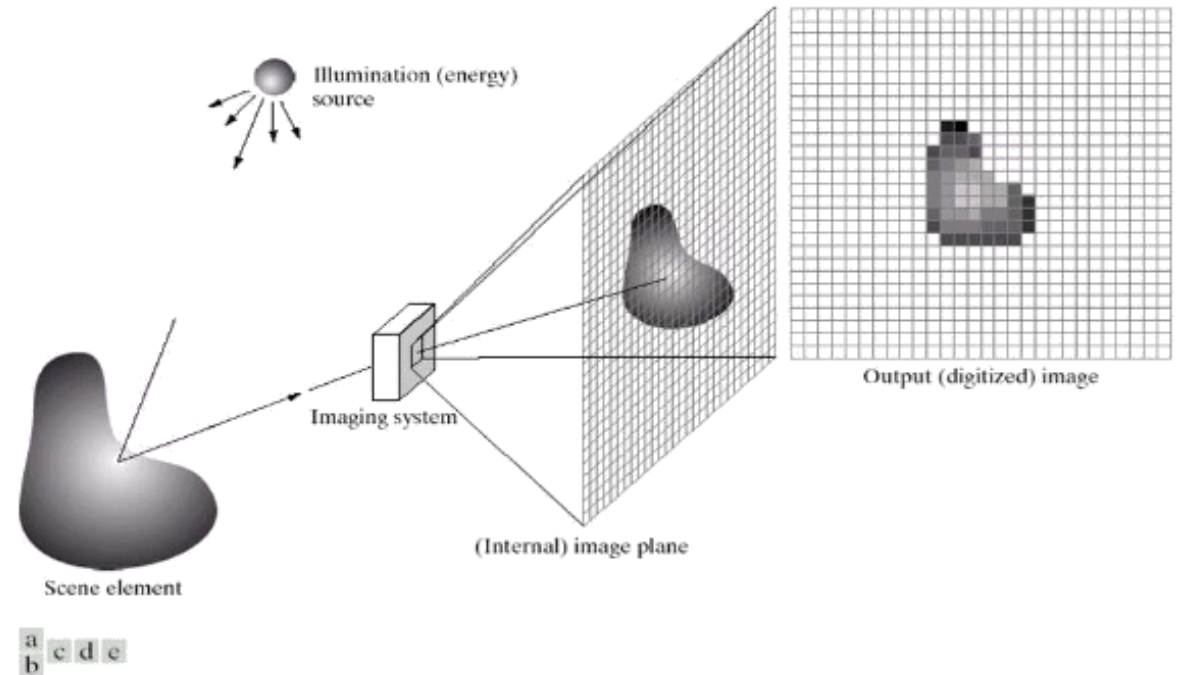
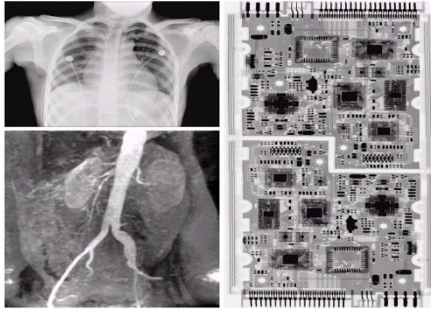


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

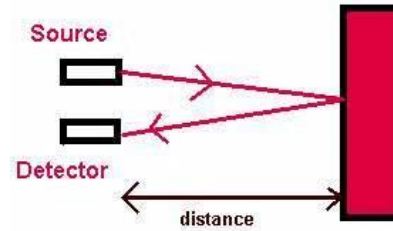
Graphic and Image: Modalities



X-ray imaging



Imaging in visible band (Photography)

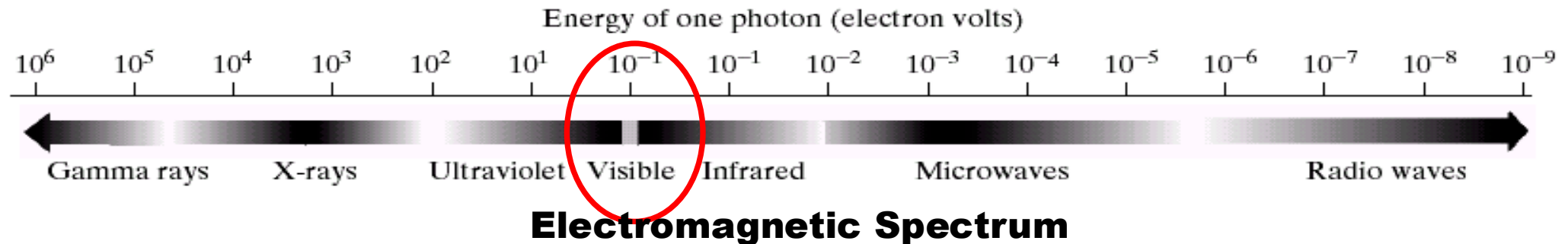


Acoustic Imaging



Computer generated Images

Electromagnetic (EM)
energy spectrum



EM Spectrum Image Examples

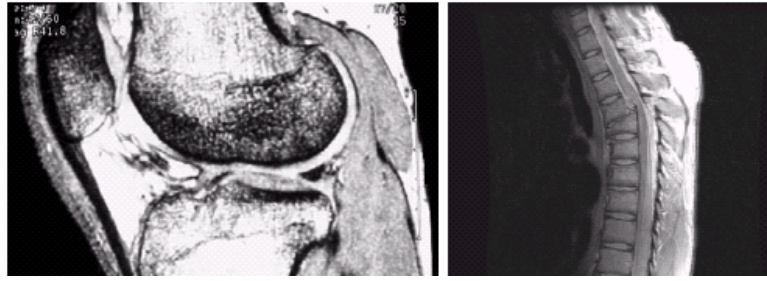


- Imaging in **visible** band
 - Photography
 - Video surveillance
 - Biometrics: fingerprint recognition



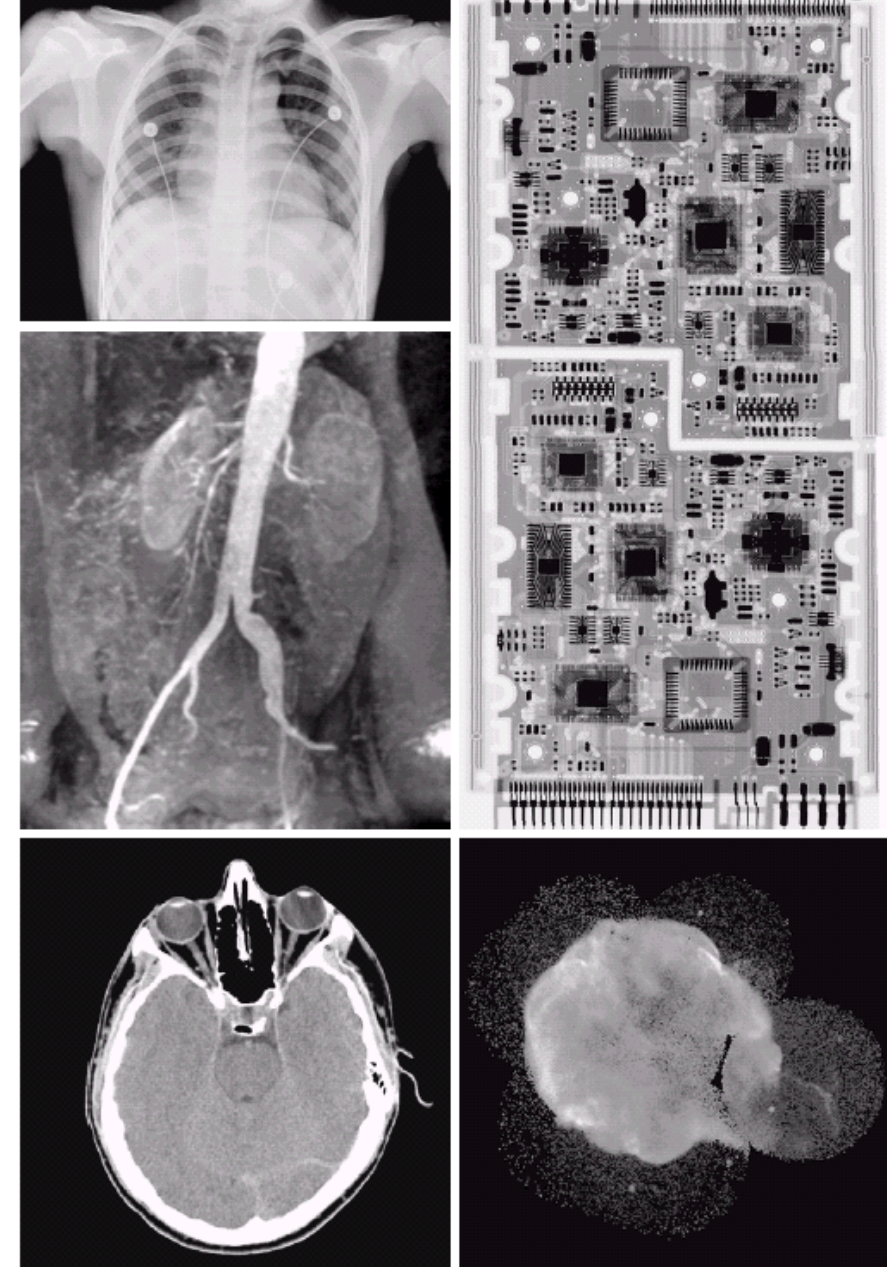
EM Spectrum Image Examples

- X-ray imaging
 - X-rays are used extensively in medical imaging and industry.
- Imaging in **radio** band
 - **Magnetic Resonance Imaging (MRI)**: The patient is placed in a powerful magnet. Radio waves are passed through his/her body in short pulses



a b

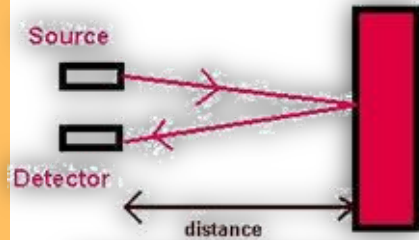
FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



a
b
c
d
e

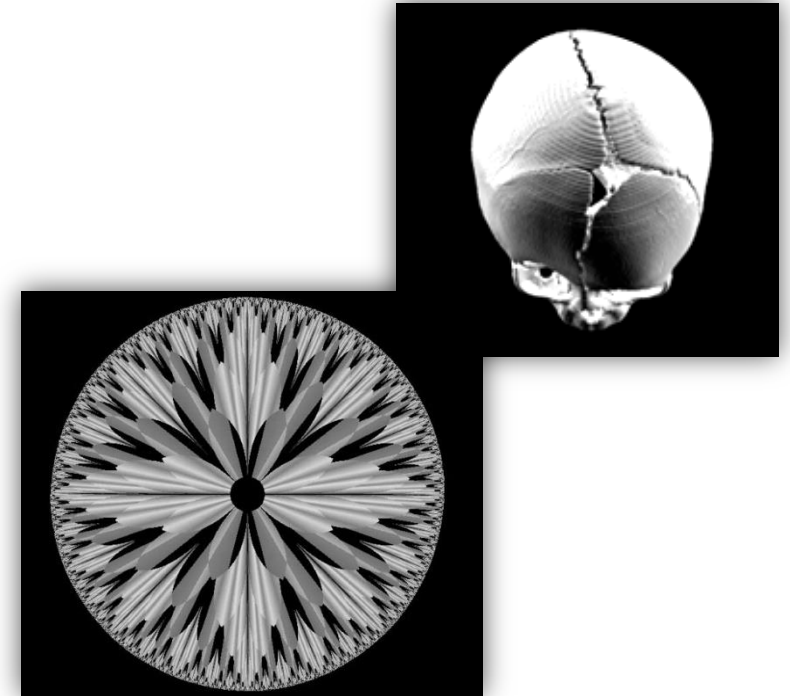
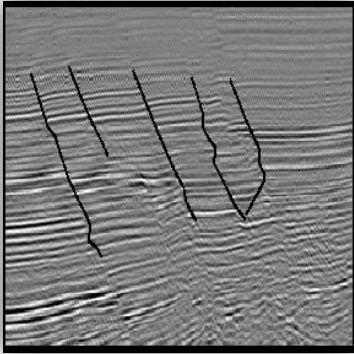
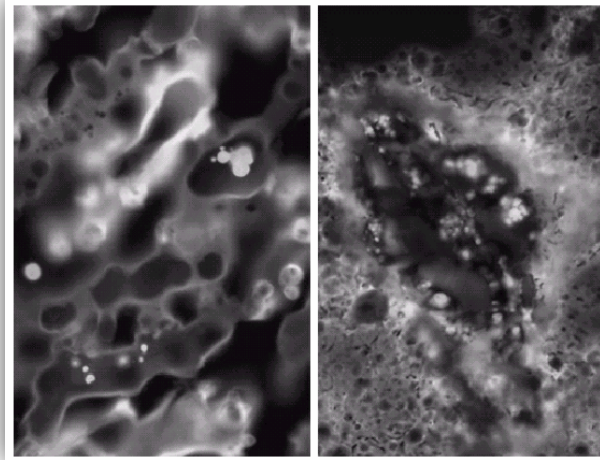
FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center, (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, (d) Mr. Joseph E. Pascente, Lixi, Inc., and (e) NASA.)

Other Imaging Modalities



normal corn

smut corn



Acoustic Imaging

- Ultrasound system transmits high-frequency (1 to 5 MHz) sound pulses into the body.
- Seismic images

Fluorescence Microscopy

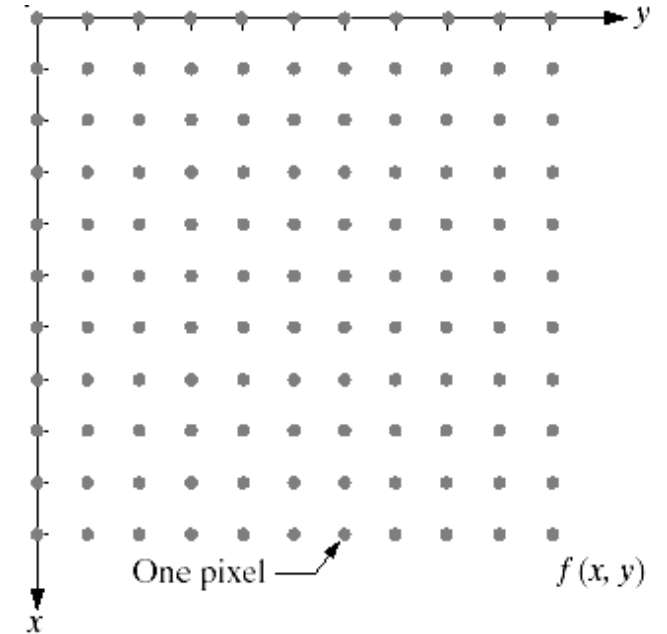
- Operate in infrared frequency

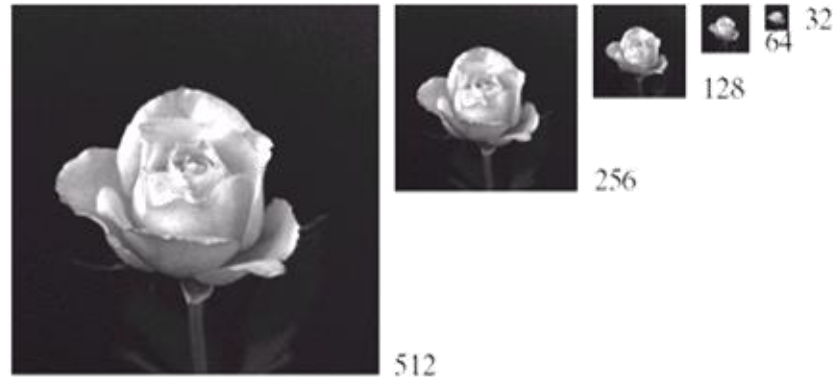
Computer generated Images

- Fractals are examples of computer-generated images.
- 3-D modeling is another example.

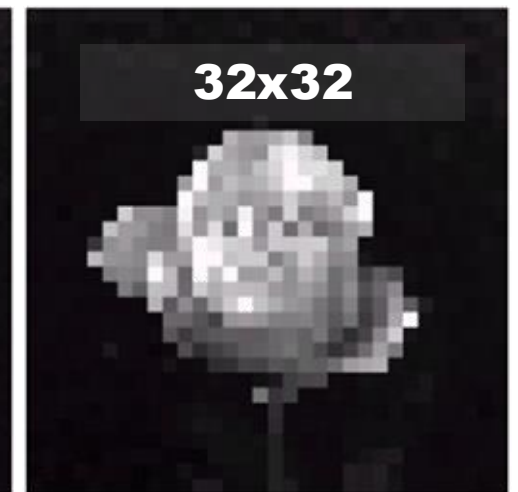
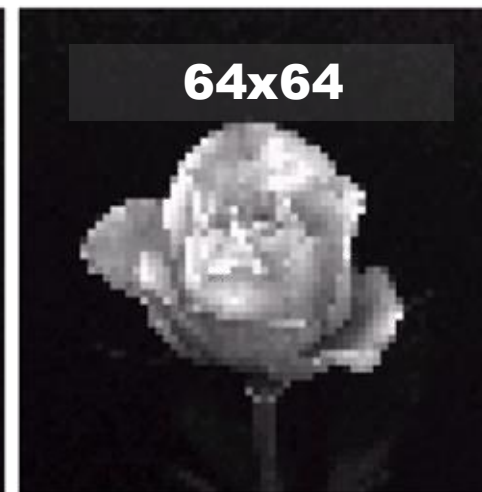
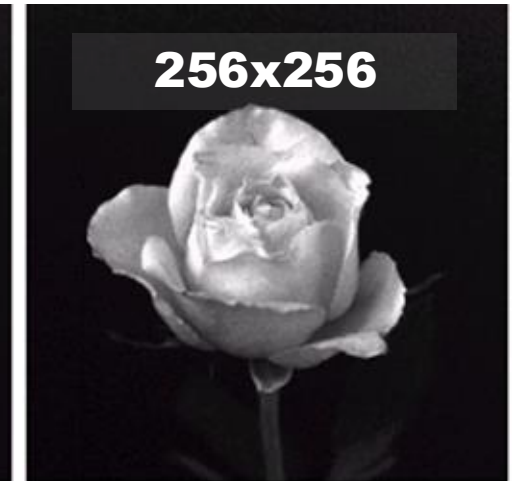
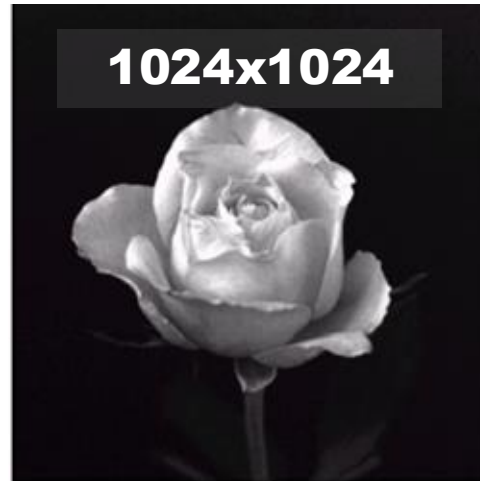
Sampling & Quantization

- Computer processing: image $f(x,y)$ must be **digitized** both spatially **and** in **amplitude**
 - **Sampling** >> digitization in spatial coordinates.
 - **Quantization** >> digitization in amplitude.
- An image will be represented as a matrix $[f(i,j)]_{N \times M}$ with $i = 1:N; j = 1:M; f \in [0, G - 1]$
 - Normally, we set $N = 2^n, M = 2^m, G = 2^k$.
- The number of bits to store the image: **$space = N \times M \times k$** .
 - The larger the values of N, M and $G = 2^k$, the better approximation of a continuous image. However, the storage and processing requirements increase as well.
 - $N \times M$ is called **spatial resolution**.
 - G is called **gray-level resolution**, k is the **bit depth**.

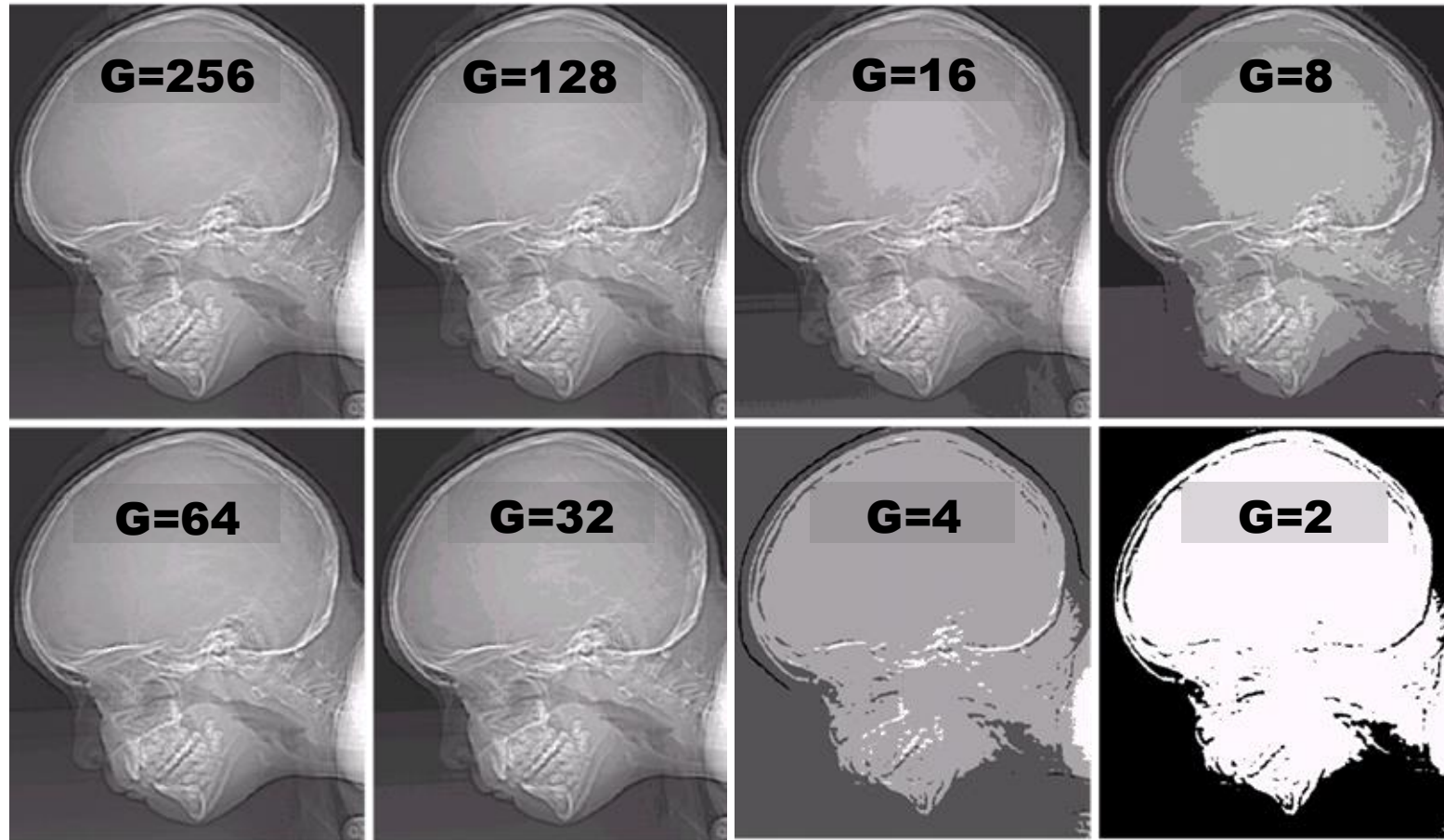




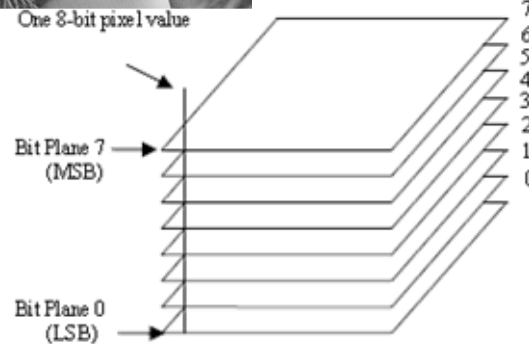
Effects of Reducing Spatial Resolution



Effects of Reducing Gray Level Resolution



- An image is typically represented as a matrix or a series of arrays.
- Each element is called a pixel.
- Bit depth is determined by the number of bits used to define each pixel.



RGB



red



green



blue

125	200	225
105	150	255
15	75	175

Bit Depth (Number of bits used to define each pixel)

1-Bit Image

1 / 0

How does a
computer store pixels

8-Bit Image (8bit=1byte)

1 0 0 0 1 0 1 0 = 138

[0, 255 ($2^8 - 1$)]

24-Bit Image

(233, 001, 138)

11101001, 00000001, 10001010

1-bit Image

- Each pixel is stored as a single bit (0 or 1), so also referred to as **binary image**.
- Such an image is also called **black-white image** because **black** is represented by 0 and **white** is represented by 1.



8-bit Lena



1-bit Lena

8-bit Image

- Each pixel has a grey-value between 0 and 255.
- Thus each pixel can be represented by a single byte, i.e. 8 bits.
 - E.g., a dark pixel might have a value of 10, and a bright one might be 230.
- 8-bit image can be thought of as a set of 1-bit **bit-planes**, where each plane consists of a 1-bit representation of the image at higher and higher levels of “elevation”.



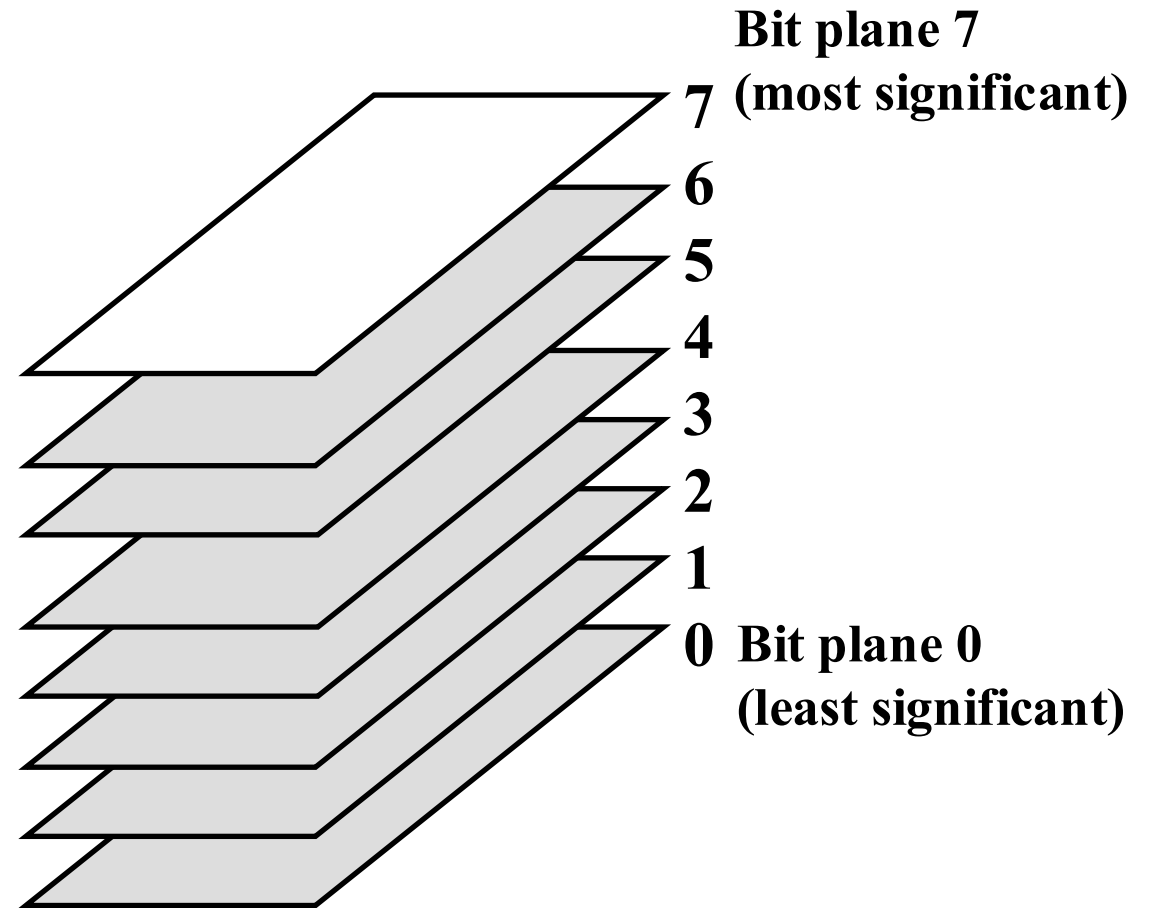
8-bit Lena



1-bit Lena

Bit-plane

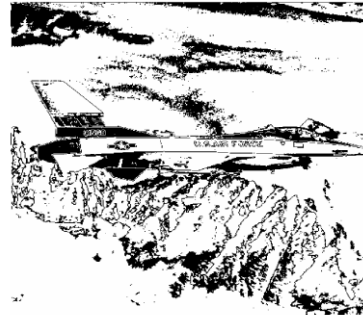
- A bit plane of a digital discrete signal (such as image or sound) is a set of bits corresponding to a given **bit position** in each of the **binary numbers** representing the signal.
- **Higher** order bit planes of an image carry a **significant** amount of visually relevant details.
- **Lower** order bit planes add **fine** (often **imperceptible**) details to the images.



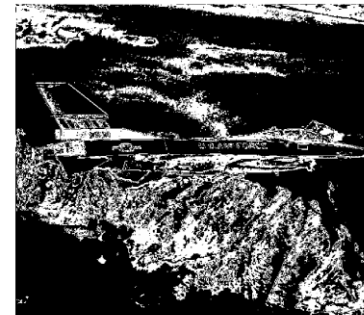
Bit-plane slicing



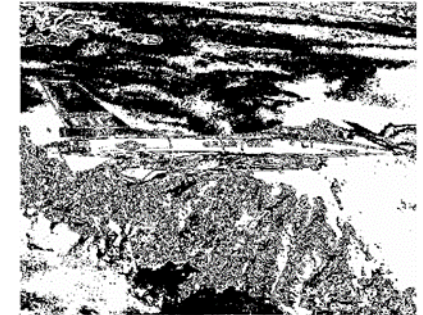
Bit plane 7



Bit plane 6



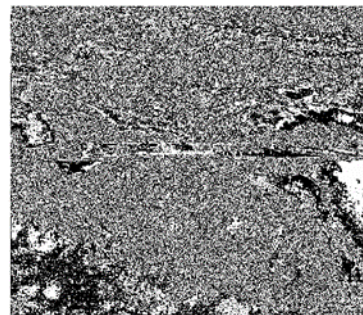
Bit plane 5



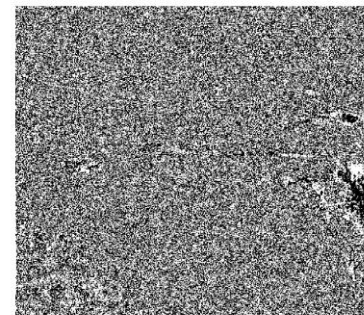
Bit plane 4



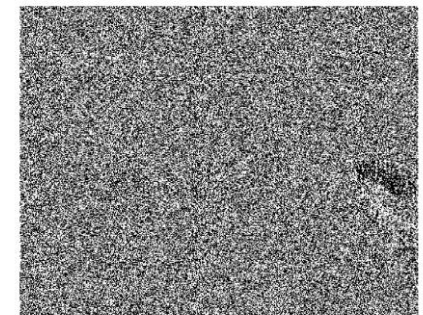
Bit plane 3



Bit plane 2



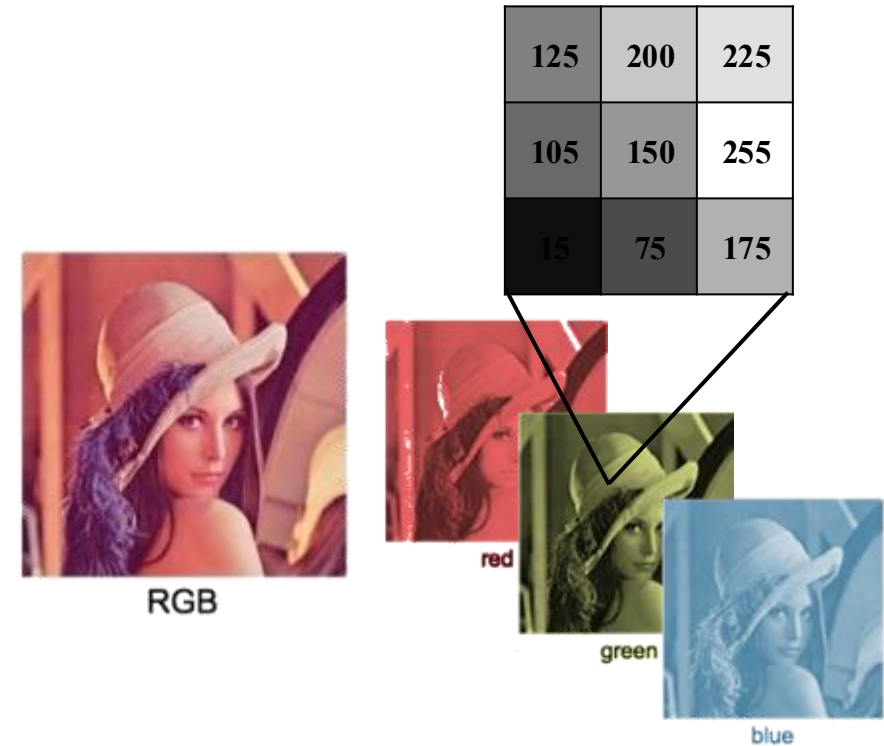
Bit plane 1



Bit plane 0

24-bit Color Image

- To represent color images, normally we use 24 bits (3 bytes) for each color. Each byte represents a color channel.
 - Supports $256 \times 256 \times 256 = 16,777,216$ possible colors.
 - However such flexibility does result in a **storage** penalty: A 640×480 24-bit color image would require 921.6 kB of storage without any compression.
- Many 24-bit color images are actually stored as **32-bit** images, with the **extra byte** of data for each pixel used to store an **alpha** value representing **special effect** information (e.g., transparency).



RGB

red

green

blue

24-Bit Image

(233, 001, 138)

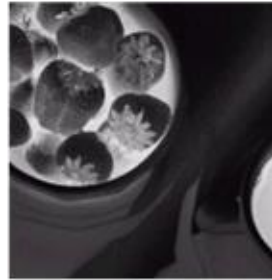
11101001, 00000001, 10001010

Color Channels

- Color digital images are made of pixels, and pixels may be represented using a color model. normally we use 24 bits (3 bytes) for each color. Each byte represents a color channel.
- RGB
- CMYK
- HSI
- YUV, YIQ
- ...



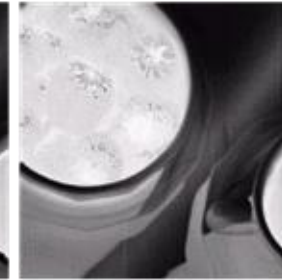
Full color



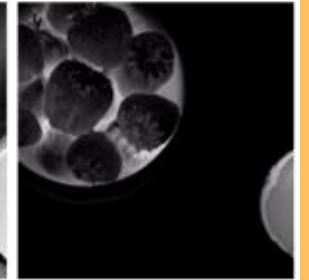
Cyan



Magenta



Yellow



Black



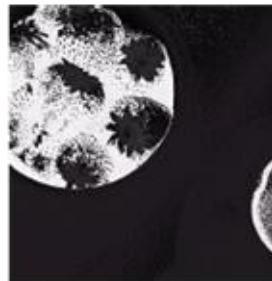
Red



Green



Blue



Hue



Saturation



Intensity

Exercise

A 4x4 image is given below. What is the **7-th bit plane** of this image?

17	64	128	128
15	63	132	133
11	60	142	140
11	60	142	138

0	0	1	1
0	0	1	1
0	0	1	1
0	0	1	1

133	10000101	1
11	00001011	0
60	00111100	0
142	10001110	1
140	10001100	1
11	00001011	0
60	00111100	0
142	10001110	1
138	10001010	1

步骤 2: 提取第 7 位, 构成 4x4 的位平面

最终第 7 位平面为:

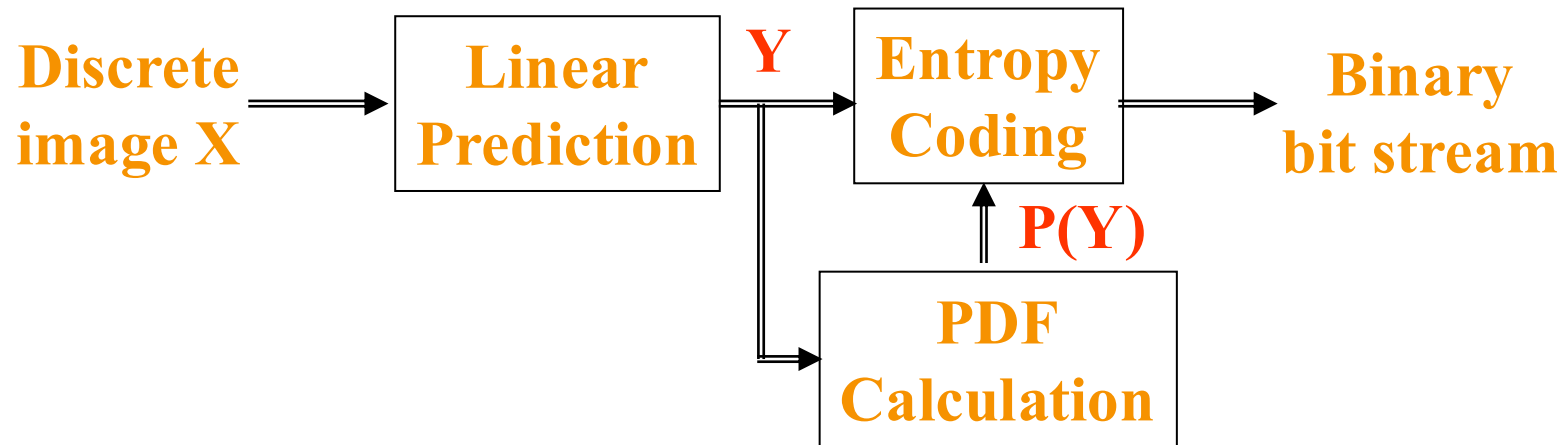
$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Lossless Image Compression

- Lossless image coding,
LPC
- JPEG-LS

Lossless Predictive Coding (LPC)

- Prediction error sequence **Y** usually contains less entropy than the original sequence **X**



Predictor

$$\hat{X}_{m,n} = \sum_{k=1}^K a_k X_k, \forall m, n$$

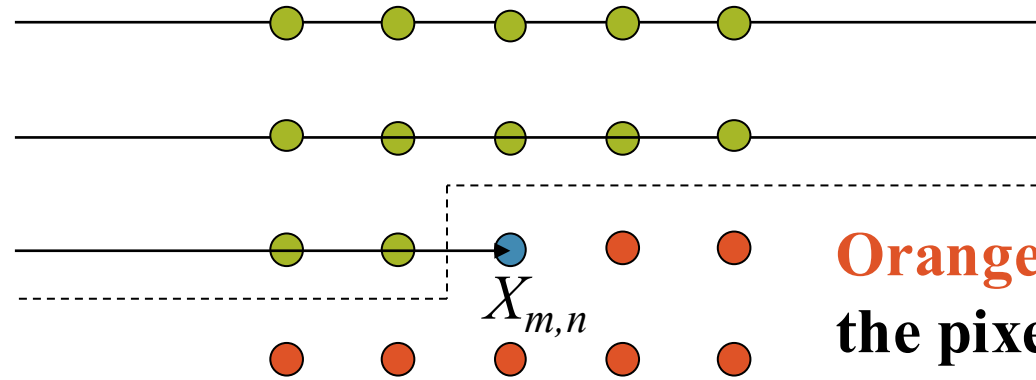
Difference

$$Y_{m,n} = X_{m,n} - \hat{X}_{m,n}$$

2D Image LPC

Raster scanning order: left $\rightarrow$ right, top $\rightarrow$ bottom

Light green
points:
available pixels



Orange points:
the pixels to be
coded

Blue point: the pixel being coded

Predictor

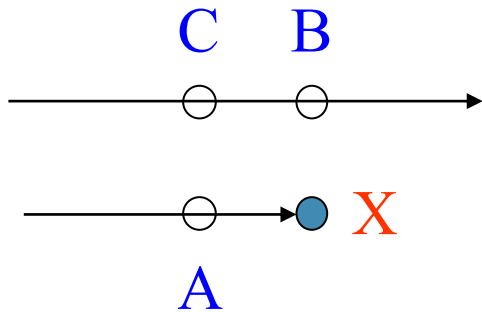
$$\hat{X}_{m,n} = \sum_{k=1}^K a_k X_k, \forall m, n$$

Difference

$$Y_{m,n} = X_{m,n} - \hat{X}_{m,n}$$

Predictors in Lossless JPEG (JPEG-LS)

Predictors in JPEG-LS



- **JPEG-LS** is a special case of JPEG image compression.
- We predict **X** using A, B and C.

P1	A (horizontal predictor)
P2	B (vertical predictor)
P3	C
P4	$A+B-C$
P5	$\text{Median}\{A,B,A+B-C\}$
P6	$\lfloor A+(B-C)/2 \rfloor$
P7	$\lfloor B+(A-C)/2 \rfloor$
P8	$\lfloor (A+B)/2 \rfloor$

Example

- Suppose we have the following image, What is the entropy of this image?

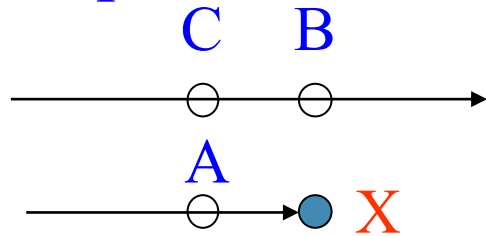
10	12	13	14
12	15	14	15
11	16	18	13
13	10	11	12

10	11	12	13	14	15	16	18
2/16	2/16	3/16	3/16	2/16	2/16	1/16	1/16

$$\eta = -\sum_i p_i \log_2 p_i = 2.9056$$

Example

- We code the image as follows
 - Code the first row using P1
 - Code the first column using P2
 - Code the other pixels using P5
- Then the **prediction error** image is



P1	A (horizontal predictor)
P2	B (vertical predictor)
P5	$\text{Median}\{A, B, A+B-C\}$

10	12	13	14
12	15	14	15
11	16	18	13
13	10	11	12

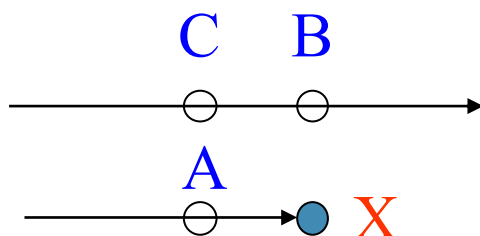
Original Image

	10	12	13
10			
12			
11			

Predict the first row
and the first column

Example

- We code the image as follows
 - Code the first row using P1
 - Code the first column using P2
 - Code the other pixels using P5
- Then the **prediction error** image is



P1	A (horizontal predictor)
P2	B (vertical predictor)
P5	Median{A,B,A+B-C}

10	12	13	14
12	15	14	15
11	16	18	13
13	10	11	12

Original Image

	10	12	13
10	12	15	14
12	14	15	18
11	16	12	11

Prediction Image

	12	15	14
	11	16	18
	13	10	11

P1 = A

	12	13	14
	15	14	15
	16	18	13

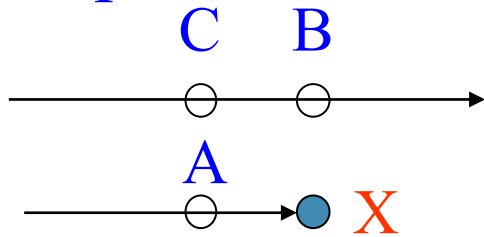
P2 = B

	14	16	15
	14	15	19
	18	12	6

A+B-C

Example

- We code the image as follows
 - Code the first row using P1
 - Code the first column using P2
 - Code the other pixels using P5
- Then the **prediction error** image is



P1	A (horizontal predictor)
P2	B (vertical predictor)
P5	Median{A,B,A+B-C}

10	12	13	14
12	15	14	15
11	16	18	13
13	10	11	12

Original Image

	10	12	13
10	12	15	14
12	14	15	18
11	16	12	11

Prediction Image

10	2	1	1
2	3	-1	1
-1	2	3	-5
2	-6	-1	1

Prediction Error Image

Example

10	12	13	14
12	15	14	15
11	16	18	13
13	10	11	12

10	2	1	1
2	3	-1	1
-1	2	3	-5
2	-6	-1	1

The **PDF** of the original image

10	11	12	13	14	15	16	18
2/16	2/16	3/16	3/16	2/16	2/16	1/16	1/16

$$\eta = - \sum_i p_i \log_2 p_i = 2.9056$$

The **PDF** of the prediction error image

1	2	3	10	-1	-5	-6
4/16	4/16	2/16	1/16	3/16	1/16	1/16

$$\eta = - \sum_i p_i \log_2 p_i = 2.5778$$

- The entropy of the prediction error is **smaller**.

Example: Lena



Vertical predictor P2
 $\eta = 4.67$ bits



Predictor P5
 $\eta = 4.55$ bits

The Next Few Lectures Will Cover

- 1. Lossy compression**
- 2. DCT and wavelet transform**

